

W1 PRACTICE

PART 1 – EXPLORATION

Learning objectives

- Apply type **inference** for variable declarations.
- Handle **nullable** and **non-nullable** variables.
- Differentiate between **final** and **const**.
- Manipulate **strings, lists, and maps**.
- Use **loops** and **conditions** to control flow.
- Define and call functions with positional and **named arguments**, understand **arrow syntax**



LO NO AI

Solve problems entirely on your own.

How to run Dart code?

You can write you code on VS Code, or using this online editor

- [Install Dart and Flutter SDK](#)
- [Online Dart Compiler](#)

Resources for this research

To help you complete this handout, use the following resources:

- | | | |
|-------------------------------|----------------------------------|------------------------------|
| ✓ Variables | ✓ Built-in types | ✓ Conditions |
| ✓ Null Safety | ✓ Lists | ✓ Functions |
| | ✓ Loops | |



EX 1 - Type Inference

EXPLAIN : Explain how Dart infers the type of a variable.

Dart infers the type of variable by guessing the value of the variable

CODE : Complete the bellow code to illustrate the concepts:

```
// Declare a int variable and let Dart infer its type
var num=12;
// Define a variable with an explicit String type
String name=Sokheng;
```

EX 2 - Nullable and Non-Nullable Variables

EXPLAIN : Explain nullable vs non-nullable variables.

Nullable variable is any variable that value can be null and non-nullable is any variable that their value cannot be null such as int variable.

EXPLAIN : When is it useful to have nullable variables?

It is useful when you want to declare a variable but don't want to put the value for it yet

CODE : Complete the bellow code to illustrate the concepts:

```
// Declare a nullable integer variable and assign it a null value
int? num=null;
// Declare a non-nullable integer variable and assign it a value
int num=12;
// Assign a new value to the nullable variable
int? num=null;
num=12;
```

EX 3 - Final and const

EXPLAIN : Describe the difference between final and const.

Final can be set only once and const variable is a compile time constant.

CODE : Complete the bellow code to illustrate the concepts:

```
// Declare a final variable and assign it the current date and time

// Can you declare this variable as const? Why?

// Declare a const variable with a integer value

// Can you reassign the value of this final variable? Why?
```

